

SEDAR 38 South Atlantic and Gulf of Mexico King Mackerel

Gulf of Mexico

Data Workshop

Life History Working Group Recommendations

1) Examine population connectivity throughout the Gulf and S. Atlantic using otolith elemental and stable isotope signatures of age-0 fish as natural tags of various regions. Otolith signatures of juvenile king mackerel collected in various resource surveys should first be examined to determine if population- or region-specific differences exist in otolith signatures, although success seems likely given the degree of classification success seen in adult mackerel whose otolith chemical signatures are integrated over several years of life, which adds greater variance to their signatures. Once otolith chemical signatures are determined, the chemistry of adult cores could be sampled to examine interregional mixing between purported migratory groups (populations) in the Atlantic, eastern Gulf, western Gulf, and even Mexico.

From SEDAR16

2) Investigate and quantify mixing between eastern Gulf and western Gulf populations using the new next-generation DNA sequencing techniques and/or otolith elemental and stable isotope analyses. The magnitude of the Mexican landings in comparison to U.S. landings from the GOM unit (annually 3-4 times higher during last 20 yr) indicates clarification of this issue should be a priority for future assessments (see SEDAR38_com_DW_Day4-2 presentation). **Modified from SEDAR16**

recommendation.

3) Further investigate/estimate the vulnerability of the western Gulf migratory group to overfished Mexican fisheries in winter (Chavez and Arreguin-Sanchez 1995). **From SEDAR16**

4) Conduct studies and monitoring that will allow estimation of natural mortality. **From SEDAR16**

5) Continue holding ageing workshops and training to standardize techniques and increase the ageing precision among laboratories. **From SEDAR16**

6) Increase age sampling in South Carolina and Georgia and length sampling north of Florida in the Atlantic. **From SEDAR16**

7) Try to recover and include age and size data from Collins et al. (1989) Atlantic age and growth study in the next stock assessment of Atlantic king mackerel. **From SEDAR16**

8) Establish clear priorities for added reproductive information as expanded work would involve considerable costs for a long-term sampling program. **From SEDAR16**

9) If made a priority, more precisely determine 1) the extent of hydration that can be determined via routine observations in the field and 2) the timing of this phase relative to final oocyte maturation and spawning and 3) calibration of the degeneration of post-ovulatory follicles. This is needed to account for and correct a likely bias in spawning frequency estimates. **From SEDAR16**

10) If made a priority, design and implement a reproductive sampling program (in concert with age sampling) on an annual basis that expands and intensifies spatial and temporal coverage (particularly adding the western Gulf of Mexico). A goal would be to provide annual estimates of spawning frequency. This would include regular training of port agents and scientific observers in macroscopic methods and additionally include a quality control component of random sub-sampling for histological comparisons. **From SEDAR16**

Commercial Fisheries Working Group Recommendations

- Consistent and sufficient levels of observers are needed in both the Gulf of Mexico and the South Atlantic. The South Atlantic shrimp fishery has especially been under sampled.
- Increase Biological Sampling efforts to better define mixing zone boundaries in the South Atlantic and Gulf of Mexico.
- Increase cooperative research with Mexican scientists to understand the relationships between king mackerel exploited in Mexican and U.S. waters. Additionally, participation of Mexican scientists is needed in the assessment process (both accumulation and interpretation of data as well as assessment) to better understand the Mexican fisheries and possible connectivity of Gulf stocks.

Recreational Fisheries Working Group Recommendations

- 1) Evaluate the technique used to apply sample weights to landings.
- 2) Develop methods to identify angler preference and targeted effort.
- 3) Continue and expand fishery dependent at sea observer surveys to collect discard information. This would help to validate self-reported headboat discard rates.
- 4) Track Texas commercial and recreational discards.
- 6) Evaluate existing and new methods to estimate historical landings

Indices of Relative Abundance Working Group Recommendations

- 1) Fisheries independent sampling continues and be expanded to the extent practical, employing consistent sampling protocols.
- 2) The defined ages that each of the recommended fishery dependent indices applies to be evaluated based on catch-at-size or catch-at-age information.
- 3) Censored regression modeling approaches (adapted from SEDAR 31) be applied to recreational fishery dependent indices of abundance to evaluate bag limit effects on catch rate indices.
- 4) Evaluation of environmental (e.g., temperature, salinity) effects on CPUE indices. The workgroup recommends that inclusion of environmental covariates that demonstrate long-term trends be

carefully considered whether the covariates are likely to affect the population abundance or the catchability of the gear. If the effect is thought to be on the population abundance, then the covariate should be excluded from the catch rate standardization and incorporated into the assessment model. If the covariate is thought to affect the catchability of the gear (e.g., fish behavior changes as temperature increases or decreases), then the covariate should be incorporated into the catch rate standardization. The strongest effects are predicted to occur during distinct periods of coldwater upwelling, as this hypothesis deserves further evaluation.

- 5) The South Carolina Pier Recreational Pier Survey was excluded from the assessment model; however, the data represent a catch record from two fixed sites. Therefore, data from this survey represent repeated measures of catch and may be useful for evaluating environmental covariates effects on catches of King mackerel.
- 6) Evaluation of the delta-lognormal generalized linear model structure. Specifically, the appropriateness of modeling factor interactions as random effects and the effect of this assumption on the resulting mean and variance estimates.
- 7) Stock assessment analysts evaluate density-dependent effects on gear catchability, to the extent possible. The hypothesis that catchability increases with the abundance of King mackerel, particularly juveniles, was proposed by stakeholders at the data workshop. It is recommended that a sensitivity run of the base assessment model include this assumption, and that this sensitivity run is compared and ranked with a base model that assumes constant catchability over time.

Assessment Workshop

1. Develop scientific survey to obtain reliable age/size composition data. This is needed, particularly as the composition data coming from the fisheries is substantially impacted by changing selectivity. This might be done with a handline survey of fixed sites. The idea would be not necessarily to get a random sample of the age composition but a reliable, relative estimate where selectivity can be assumed constant. An index would be beneficial.
2. Evaluate environmental influence on recruitment, larval/juvenile survival
3. Determine stock mixing rates using genetic methods, otolith microchemistry or otolith shape.
4. Develop/Evaluate methods to maintain continuity of fishery-dependent indices in light of management regulations and ITQs.
5. Determine most appropriate methods to deal with changing selectivity in fisheries over time, particularly changing selectivity related to management actions or targeting of specific cohorts.
6. Evaluate most appropriate methods to deal with unreliable historic discard size composition data so that discard ratios can be reliably estimated.
7. Research on U.S. Gulf of Mexico stock overlap with King Mackerel landed by Mexico is needed.

Review Workshop

1. Develop a scientific survey to obtain reliable age/size composition data. This is needed, particularly as the composition data coming from the fisheries is substantially impacted by changing selectivity. This might be done with a handline survey of fixed sites. The idea would be not necessarily to get a random sample of the age composition but a reliable, relative estimate where selectivity can be assumed constant. An index would be beneficial. The review panel recommends that the design of a scientific survey be peer reviewed.
2. Determine most appropriate methods to deal with changing selectivity in fisheries over time, particularly changing selectivity related to management actions or targeting of specific cohorts. The review panel suggests that historical mark-recapture data available from NMFS SEFSC (Panama City) and FWRI could be used to compare size composition of recaptures for different fishing gears to evaluate selectivity for historic periods.
3. Conduct research on the U.S. Gulf of Mexico stock overlap with Mexico. The review panel recommends this work include determination of mixing rates/connectivity between the eastern and western Gulf migratory groups using otolith shape and/or microchemistry analysis, as well as model simulations to evaluate the impact of Mexican harvest on the putative single Gulf of Mexico stock.
4. Determine stock mixing rates using otolith microchemistry and/or otolith shape analysis on a routine basis that would allow future stock assessments to capture the dynamic spatial and temporal nature of mixing of the Atlantic and Gulf of Mexico stocks, and consider evaluating stock mixing within integrated modeling approaches.
5. Quantify tournament landings from the Gulf of Mexico.
6. Develop/Evaluate methods to maintain continuity of fishery-dependent indices in light of management regulations and ITQs.
7. Consider conducting an extensive tagging program to: a) better understand migration patterns; b) provide additional and individual growth rate information; c) better understand fishery selectivity; d) provide fishery exploitation rates; e) provide information about natural mortality rates. Fishery independent recapture information (i.e. use acoustic and satellite tags) will assist with a). Age at capture information of tagged animals will assist with b). A multi-year tagging program will be required for e). The review panel recommends that a specific workshop be held to consider in detail the design of a tagging program.

South Atlantic

Data Workshop

Life History Working Group Recommendations

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- 11) Evaluation of environmental (e.g., temperature, salinity) effects on CPUE indices. The workgroup recommends that inclusion of environmental covariates that demonstrate long-term trends be carefully considered whether the covariates are likely to affect the population abundance or the catchability of the gear. If the effect is thought to be on the population abundance, then the covariate should be excluded from the catch rate standardization and incorporated into the assessment model. If the covariate is thought to affect the catchability of the gear (e.g., fish behavior changes as temperature increases or decreases), then the covariate should be incorporated into the catch rate standardization. The strongest effects are predicted to occur during distinct periods of coldwater upwelling, as this hypothesis deserves further evaluation.

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- 13) Evaluation of the delta-lognormal generalized linear model structure. Specifically, the appropriateness of modeling factor interactions as random effects and the effect of this assumption on the resulting mean and variance estimates.
- 14) Stock assessment analysts evaluate density-dependent effects on gear catchability, to the extent possible. The hypothesis that catchability increases with the abundance of King mackerel, particularly juveniles, was proposed by stakeholders at the data workshop. It is recommended that a sensitivity run of the base assessment model include this assumption, and that this sensitivity run is compared and ranked with a base model that assumes constant catchability over time.

Assessment Workshop

1. Evaluate environmental influences on recruitment and larval/juvenile survival, focusing on potential predator prey impacts, hydrodynamic influences on recruitment, pollution, HABs or excess nutrient run-off.
2. Develop scientific survey to obtain reliable age/size composition data and relative abundance of adult fish. This could be done using gillnets or handlines.
3. Determine dynamic stock mixing rates using genetic methods, otolith microchemistry, stable isotopes or otolith shape. Even though the mixing zone is now smaller, stock mixing is a dynamic process and may vary substantially from one year to the next, particularly in relation to cold temperature years that may compress both the Gulf and Atlantic stock into South Florida waters.
4. Evaluate the stock mixing within integrated modeling approaches
5. Expand SEAMAP trawl survey below the Cape Canaveral area and potentially into deeper shelf waters
6. Quantify tournament landings from the Gulf of Mexico
7. Determine if female spawning periodicity varies by size or age.
8. More accurately characterize juvenile growth by increasing samples of age-0 and 1 fish.
9. Conduct studies to estimate of natural mortality.

Review Workshop

1. Develop a survey to obtain reliable age/size composition data and relative abundance of adult fish. This could be done using gillnets or handlines. The review panel recommends that the design of a scientific survey be peer reviewed.
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3. Determine stock mixing rates using otolith microchemistry and/or otolith shape analysis on a routine basis that would allow future stock assessments to capture the dynamic spatial and temporal nature of mixing of the Atlantic and Gulf of Mexico stocks, and consider evaluating stock mixing within integrated modeling approaches.
4. More accurately characterize juvenile growth by increasing samples of age-0 and 1 fish. Further investigate 2-phase growth models including different breakpoints and different growth models to better model size and age. Consider if there is temporal (annual and seasonal) variability in growth rates. Results of this analysis in terms of the best model will need to be implementable in SS3 to continue with the integrated modeling approach.
5. Determine if female spawning periodicity varies by size or age.
6. Expand the SEAMAP trawl survey below the Cape Canaveral area and potentially into deeper continental shelf waters.
7. Consider conducting an extensive tagging program to: a) better understand migration patterns; b) provide additional and individual growth rate information; c) better understand fishery selectivity; d) provide fishery exploitation rates; and e) provide information about natural mortality rates. Fishery independent recapture information (i.e., use acoustic and satellite tags) will assist with a). Age at capture information of tagged animals will assist with b). A multi-year tagging program will be required for e). The review panel recommends that a specific workshop be held to consider in detail the design of a tagging program.